

Name	()	Class	
------	----------	-------	--



RIVER VALLEY HIGH SCHOOL
2018 Year 6 Preliminary Examination
Higher 2

MATHEMATICS

9758/01

Paper 1

13 September 2018

3 hours

Additional Materials: Answer Paper
List of Formulae (MF26)
Cover Page

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, class and index number in the space at the top of this page.

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphic calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

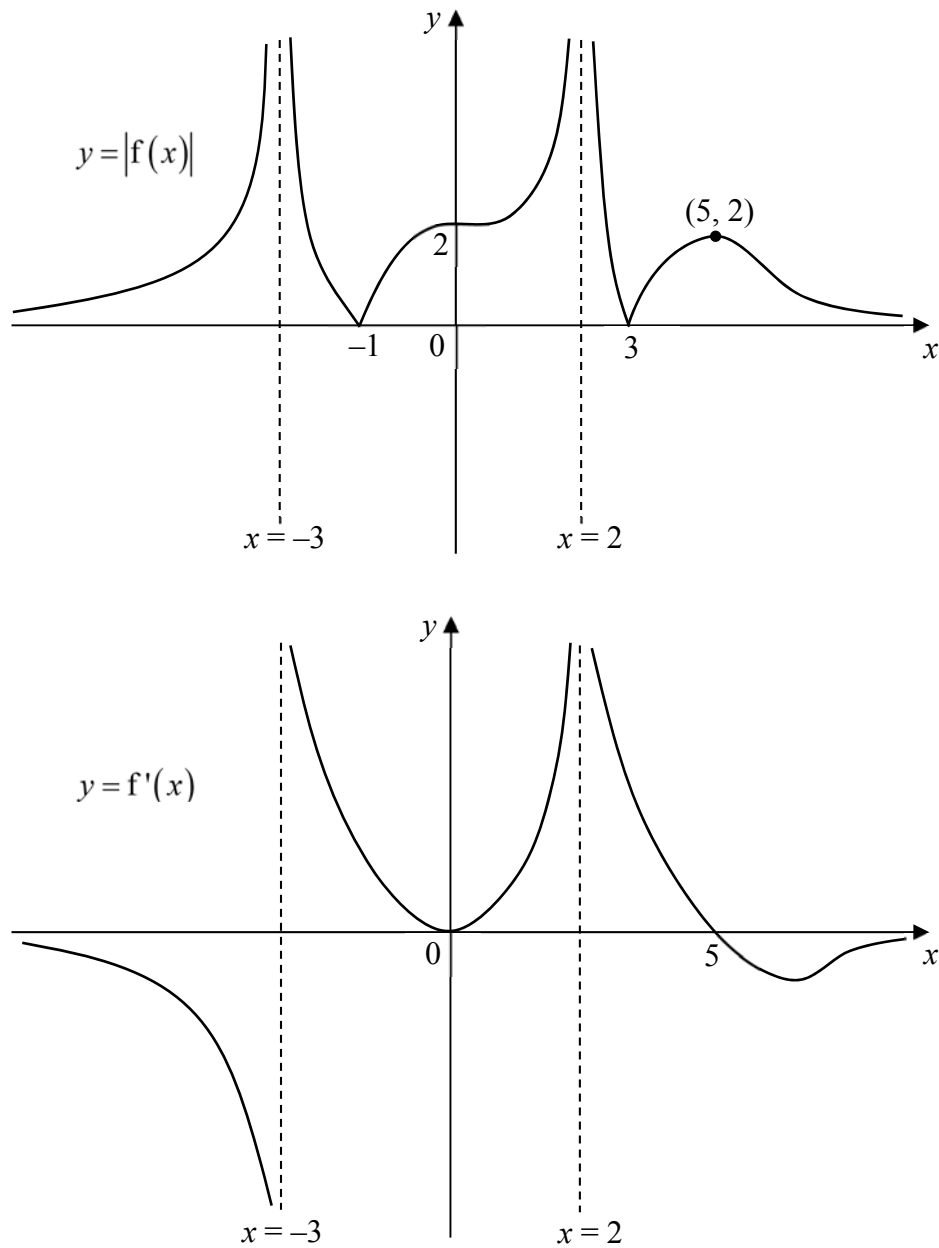
This document consists of **8** printed pages.

1. Relative to the origin O , the points A , B , C and D have position vectors

$$\overrightarrow{OA} = \begin{pmatrix} p \\ q \\ r \end{pmatrix}, \overrightarrow{OB} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \overrightarrow{OC} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \text{ and } \overrightarrow{OD} = \begin{pmatrix} 0 \\ -3 \\ 0 \end{pmatrix} \text{ respectively.}$$

The vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are perpendicular to each other. The vector $\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$ is a normal to the plane containing O , A and C . The point A lies on the line passing through the point D parallel to the vector $\overrightarrow{OB}$. Find the values of p , q and r . [6]

2. The diagrams show the graphs of $y = |f(x)|$ and $y = f'(x)$.



The graph of $y = |f(x)|$ has stationary points at $(0, 2)$ and $(5, 2)$, and the equations of asymptotes are $x = -3$ and $x = 2$. The equations of asymptotes of $y = f'(x)$ are $x = -3$ and $x = 2$.

On separate diagrams, sketch the graphs of

(i) $y = f(x)$; [3]

(ii) $y = \frac{1}{f'(x)}$. [3]

3. In this question, all physical quantities are of appropriate units which you need not state.

In a science experiment, charged particles from a fixed position O are bombarded towards a charged conductor. The electric potential energy (E) of each particle is given by

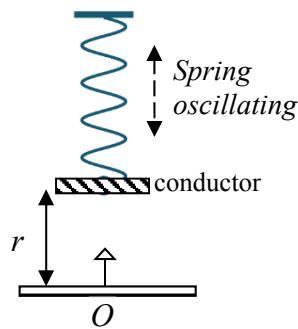
$$E = \frac{0.9qQ}{r}$$

where q is the charge of the particle, Q is the charge of conductor and r is the distance between them.

- (i) Assuming q and Q are constants, express $\frac{dE}{dt}$ in terms of q , Q , r and $\frac{dr}{dt}$. [1]

It is further given that the conductor is attached to a spring that moves in an oscillating manner such that the distance between the conductor and O is given by

$r = 0.8 + 0.6 \cos\left(\frac{1}{2}t\right)$ as shown in the diagram below.



- (ii) State the range of values of r . [1]

At a certain instant, a particle with a constant charge of 1.5 is of distance 1.1 units from the conductor, which has a constant charge of 4.0.

- (iii) Find the exact time of this instant, and the rate at which the electrical potential energy of the particle is increasing at this instant. [5]

4. Relative to the origin O , the points A , B and C are such that $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ and $\overrightarrow{OC} = \mathbf{c}$ respectively, where $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are vectors which are mutually non-parallel.

The plane Π and the line l have the following equations

$$\Pi : \mathbf{r} = \lambda \mathbf{a} + \mu \mathbf{b} \quad \text{where } \lambda, \mu \in \mathbb{R} \quad \text{and}$$

$$l : \mathbf{r} = \mathbf{a} + \beta \mathbf{c} \quad \text{where } \beta \in \mathbb{R}$$

respectively.

- (i) Find the point(s) of intersection between Π and l given that the points O , A , B and C are
- (a) coplanar,
- (b) not coplanar. [3]
- (ii) State the geometrical meaning of $\frac{1}{2}|\mathbf{a} \times \mathbf{b}|$. [1]

In the rest of the question, O , A , B and C are not coplanar.

- (iii) The vector $\mathbf{p}$ is a unit vector in the direction of $\mathbf{a} \times \mathbf{b}$. State the geometrical meaning of $|\mathbf{c} \cdot \mathbf{p}|$. [1]
- (iv) It is given that $\overrightarrow{OA} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ and $\overrightarrow{OC} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$. Find the volume of the pyramid $OABC$. [3]

5. (i) By using the substitution $x = 2 \sin^2 \theta$, find the exact value of $\int_0^1 \sqrt{2x - x^2} \, dx$. [5]
- (ii) Hence, find the exact value of $\int_0^1 \frac{x(1-x)}{\sqrt{2x-x^2}} \, dx$. [3]

6. A curve C has parametric equations

$$x = 1 - \sin 2t, \quad y = \cos t, \quad \text{for } 0 \leq t < 2\pi.$$

- (i) Find $\frac{dy}{dx}$ in terms of t . [2]
- (ii) Find the stationary points of C . [2]
- (iii) There are four points where the normals to C are parallel to the x -axis. Deduce that the four points form a rectangle. State the length and breadth of the rectangle. [5]

7. Given that $y = f(x)$ where $\sqrt{1-x^2} \frac{dy}{dx} = \frac{y}{2}$ and $f(0) = 1$.

- (i) Find the Maclaurin series for y up to and including the term in x^3 . [2]
- (ii) The Maclaurin series for $e^{y'}$, up to and including the term in x , is $p + qx$, where y' denotes the first derivative of y . Find the exact values of p and q . [4]

8. (a) Given that $z^* = \frac{(1+ai)^3}{(1-i)^2}$, where a is a positive real number.

- (i) Find a such that $|z| = 4$. [2]
- (ii) Without the use of a calculator, express z in the form $re^{i\theta}$, where r and θ are exact values, $r > 0$ and $-\pi < \theta \leq \pi$. [3]

- (b) Given that $f(z) = z^4 + 2z^3 + (\pi^2 + e^2)z^2 + pz + q$, where $p \geq 0$ and $q \geq 0$, and that $i\pi$ satisfies the equation $f(z) = 0$.

- (i) Find the exact values of p and q . [2]
- (ii) Hence find the remaining roots of $f(z) = 0$ in the form $\alpha + \beta i$, where α and β are exact real numbers. [4]

9. A geometric series G has common ratio r and an arithmetic series A has first term a and common difference d , where a and d are non-zero. The first three terms of G are equal to the third, fourth and sixth terms of A respectively. The sum of the first four terms of A is -6 .

- (i) Show that $r^2 - 3r + 2 = 0$ and find the value of r . [2]
- (ii) Deduce the values of a and d . [4]
- (iii) Determine the least value of m such that the difference between the m^{th} terms of G and A is more than 10000. [2]

Let a_n denote the n^{th} term of A . The sequence x_n is defined as $x_n = (2k)^{a_n}$ where k is a constant.

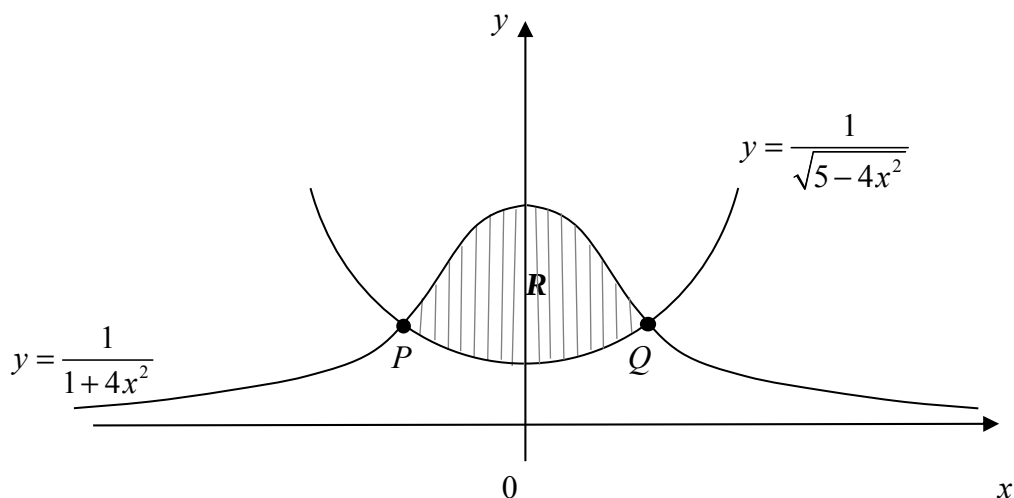
- (iv) Find the range of values of k such that the series $\sum_{n=1}^{\infty} x_n$ converges. [3]

10. A group of scientists have recently discovered the growth of a rare species of plant in a wildlife reserve area in central Africa that is well known for its severe climate changes at times. The scientists also discovered that the species of plant can be used in the cure for several human terminal illnesses.

Let N denote the number of the species of plant at time t years after its discovery.

- (i) Based on studies on other similar species of plant in other areas, the scientists believe that the rate of increase of N is proportional to the product of N and $(1000 - N)$. Given that they expect the number of plants to increase at a rate of 125 per year when there are 500 plants in the area, deduce a differential equation relating N and t . [2]
- (ii) Given further that there are initially 200 of this rare species of plant in the area, find an expression for N in terms of t and sketch of the graph of N against t . [7]
- (iii) The scientists target to have the number of the species of plants in the area to exceed 1200 by the end of 5 years. Determine if this target can be achieved. [1]
- (iv) Based on the context given, provide two reasons why the proposed model of growth for this rare species of plant may not be suitable. [2]

11. Alice is tasked to design a new logo for a toy company. The logo consists of the finite region R bounded by the curves $y = \frac{1}{1+4x^2}$, $y = \frac{1}{\sqrt{5-4x^2}}$ and between the points P and Q as shown below.



- (i) Find the x -coordinates of P and Q and show that the exact value of the area of R is given by

$$\frac{\pi}{4} - \sin^{-1}\left(\frac{1}{\sqrt{5}}\right). \quad [6]$$

Alice then decides to explore different regions bounded by various transformations of the above two curves.

- (ii) Without evaluating the actual integrals, deduce the exact areas of the finite regions bounded by the following curves. Justify your answers.

(a) $y = \frac{1}{4x^2 - 8x + 5}$ and $y = \frac{1}{\sqrt{1 + 8x - 4x^2}}$ [2]

(b) $y = \frac{2}{1 + 4x^2}$ and $y = \frac{1}{\sqrt{\frac{5}{4} - x^2}}$ [2]

In designing a related 3-dimensional logo for the company, Alice considers rotating the above finite region R about the y -axis.

- (iii) Find the volume of the 3-dimensional logo formed by rotating R 180° about the y -axis. [3]

END OF PAPER